

A Hybrid Quantum-Classical Model For Detecting & Mitigating Large Language Model Hallucinations

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Recent advancements in Large Language Models (LLMs) have perpetuated hallucinated responses/misinformation during human-AI interactions. These hallucinations largely stem from the lack of access to the internal probabilistic systems, also known as black box problems, that AI models run on. Although current research has established numerous detection frameworks, the methodologies are not determinant of misinformation because they mostly rely on theories and probabilities. In this study, we propose a hybrid classical-quantum framework that combines black-box behavioral probing (semantic entropy and softmax probability features: entropy, confidence gap and low probability) with quantum classification components for noisy intermediate-scale quantum (NISQ) hardware. We first built a classical Support Vector Machine model trained on these behavioral and confidence features, extracted from open-source LLM-generated responses utilizing hallucination benchmark datasets to evaluate whether such black-box signals can aid in separation of hallucinated outputs and factual outputs. This classical baseline was further integrated with a 4-qubit quantum classifier embodying a projected quantum kernel operation on PCA-compressed BERT embeddings and a classical RBF kernel SVM, ultimately proposing a comprehensive framework for a Hybrid Classical-Quantum LLM. Due to NISQ hardware limitations, we break down the framework to its sub-components and provide proof-of-concept validation for each via hallucination detection via confidence probing, Quantum-Inspired KV Cache compression and establishing a working pipeline of projected quantum kernel classification that requires real labelled dataset in future.

Keywords: Large Language Models, Quantum Large Language Models, Hallucinations, Classical Support Vector Machine, Confidence Probing, Quantum-inspired Compression, Projected Quantum Kernel Classification